Simple Baselines for Few-Shot Fine-Grained Recognition: Are Modern Pretrained Features Enough?

***Abstract*—Few-shot fine-grained image classification requires distinguishing visually similar categories from very few labeled examples. While recent meta-learning methods address this through complex episodic training, it remains unclear whether such complexity is necessary when modern pretrained visual representations are available. In this work, we systematically evaluate simple baseline approaches that combine frozen pretrained feature extractors (DINOv2, CLIP, ViT-B/16-IN21K, and ResNet-50) with lightweight classifiers (Logistic Regression, Nearest Class Centroid, and k-Nearest Neighbors) on three fine-grained benchmarks: CUB-200-2011, Oxford Flowers-102, and FGVC-Aircraft. Under a standard 5-way evaluation protocol with 600 episodes per configuration, our best simple baseline achieves 98.85% on CUB-200-2011 (5-shot), 99.97% on Oxford Flowers-102 (5-shot), and 82.92% on FGVC-Aircraft (5-shot), outperforming published meta-learning methods such as ProtoNet, FEAT, and DeepEMD by approximately 10 percentage points on CUB-200-2011 — with zero task-specific training. These results demonstrate that modern foundation model representations, combined with simple classifiers, provide strong and computationally efficient baselines for few-shot fine-grained recognition, and suggest that the field should re-evaluate the necessity of complex meta-learning architectures.**

***Index Terms*—few-shot learning, fine-grained recognition, image classification, pretrained representations, vision transformers, DINOv2, CLIP**

I. Introduction

Deep learning models have achieved remarkable progress in image recognition. However, the majority of deep learning systems depend heavily on large annotated datasets, which are often expensive and time-consuming to create [1], [2]. Few-shot learning addresses this limitation by enabling models to recognize novel categories from only a handful of labeled examples [1]–[3].

The challenge is compounded in fine-grained image recognition, where visual differences between classes are subtle — such as distinguishing bird species by plumage patterns, aircraft models by fuselage dimensions, or flower types by petal shape [4], [5]. To address few-shot learning, researchers have developed metric-based methods (e.g., Prototypical Networks [7] and Matching Networks [6]) that construct feature spaces where same-class samples cluster together, as well as optimization-based meta-learning methods (e.g., MAML [8]) that learn initializations enabling rapid adaptation. More sophisticated approaches such as DeepEMD [9] and FEAT [10] further improve performance through structured feature matching and task-adaptive embeddings.

While these methods achieve strong results, they employ complex architectures and elaborate episodic training procedures. Meanwhile, recent advances in large-scale visual representation learning have produced foundation models — including CLIP [12] and DINOv2 [13] — that generate highly transferable visual embeddings across diverse tasks [11]. This raises a fundamental question: *are complex meta-learning architectures still necessary for few-shot fine-grained recognition, or have modern pretrained representations rendered them unnecessary?*

In this work, we investigate whether frozen pretrained representations combined with simple off-the-shelf classifiers can match or exceed the performance of dedicated meta-learning systems on fine-grained few-shot classification tasks. We conduct a systematic evaluation across four pretrained backbones, three classifiers, three fine-grained datasets, and four shot settings, totaling 144 configurations and 86,400 episodes.

Our main contributions are as follows:

- We provide the first systematic comparison of four modern pretrained backbones (DINOv2, CLIP, ViT-B/16-IN21K, ResNet-50) for few-shot fine-grained recognition across three challenging benchmarks.
- We demonstrate that frozen pretrained features with simple classifiers achieve 98.85% accuracy on CUB-200-2011 (5-way 5-shot), outperforming state-of-the-art meta-learning methods by approximately 10 percentage points without any task-specific training.
- We reveal that the optimal backbone depends on the visual domain: self-supervised DINOv2 excels on natural categories (birds, flowers), while language-supervised CLIP is superior on manufactured objects (aircraft).
- We provide feature space analysis, failure case characterization, and practicality comparisons that offer insights into when and why simple baselines succeed or fail.

II. Related Work

A. Few-Shot Learning

Few-shot learning aims to recognize novel categories from very few labeled examples. The field has developed along two main directions. Metric-based approaches learn embedding spaces where samples from the same class cluster together. Matching Networks [6] use attention-based comparison between support and query samples. Prototypical Networks [7]

classify queries by Euclidean distance to class prototypes (mean embeddings). These methods are simple and effective but rely on the quality of the learned embedding space.

Optimization-based methods take a different approach. MAML [8] learns model initializations that enable rapid adaptation to new tasks through a few gradient steps. Meta-Baseline [11] demonstrated that standard pretraining followed by simple meta-learning can be surprisingly competitive with more complex approaches.

More recent methods introduce additional sophistication. DeepEMD [9] uses the Earth Mover's Distance for structured image comparison, while FEAT [10] introduces task-adaptive feature embeddings. While these methods achieve progressively better results, they require complex episodic training procedures and specialized architectures.

B. Fine-Grained Visual Recognition

Fine-grained recognition addresses the challenge of distinguishing visually similar subcategories within a broader class [4]. Standard benchmarks include CUB-200-2011 for bird species [5], Oxford Flowers-102 for flower types [16], and FGVC-Aircraft for aircraft variants [17]. The key difficulty lies in capturing subtle inter-class differences (e.g., plumage coloration, petal shape, fuselage length) while remaining invariant to substantial intra-class variation in pose, lighting, and background.

The intersection of few-shot learning and fine-grained recognition is particularly challenging because the subtle discriminative cues must be captured from very few examples, making it unclear whether general-purpose representations suffice or task-specific learning is essential.

C. Vision Foundation Models

Recent large-scale pretrained models have transformed visual representation learning. CLIP [12] learns visual representations aligned with natural language through contrastive pretraining on 400 million image-text pairs from the web. DINOv2 [13] employs self-supervised self-distillation on LVD-142M (142 million curated images), producing features that achieve strong performance across diverse visual tasks without any labeled data during training.

Both models build upon the Vision Transformer (ViT) architecture [14], which replaces convolutional inductive biases with self-attention mechanisms. Earlier supervised models, including ViT pretrained on ImageNet-21K and ResNet-50 [15] pretrained on ImageNet-1K, provide natural comparison points representing different scales of supervised pretraining.

D. Simple Baselines for Few-Shot Learning

Chen et al. [3] provided an influential study demonstrating that simple baselines with proper training can match sophisticated meta-learning methods, establishing that the gap between complex and simple approaches was smaller than previously believed. Their work used relatively small backbone networks (Conv-4, ResNet-12) and showed that standard classification pretraining followed by nearest-centroid evaluation could be competitive.

Our work extends this line of inquiry by asking whether *any* task-specific training is necessary: we evaluate completely frozen modern foundation models with off-the-shelf classifiers, demonstrating that representation quality alone can surpass dedicated meta-learning systems on fine-grained benchmarks.

III. Methodology

A. Overview

Our approach is deliberately simple. The pipeline consists of three steps:

1) Pass each image through a frozen pretrained backbone to obtain a feature vector.
2) L2-normalize the resulting feature vectors.
3) Train a lightweight classifier on the support set features and evaluate on the query set.

No backbone parameters are updated, no episodic training is performed, and no task-specific modules are introduced. This design isolates the contribution of the pretrained representation from the classification algorithm, directly testing whether modern visual features are sufficient for few-shot fine-grained recognition.

B. Datasets

We evaluate on three widely used fine-grained visual recognition benchmarks spanning diverse visual domains:

CUB-200-2011 [5]: Contains 11,788 images across 200 bird species. It is one of the most widely used benchmarks for fine-grained classification, requiring discrimination of subtle plumage patterns, beak shapes, and body proportions.

Oxford Flowers-102 [16]: Comprises 8,189 images of 102 flower categories collected under varying conditions of scale, pose, and illumination. Flowers are distinguished by petal shape, color patterns, and structural features.

FGVC-Aircraft [17]: Contains approximately 10,000 images of 100 aircraft model variants. This dataset is particularly challenging as visual differences between variants (e.g., Boeing 737-700 vs. Boeing 737-800) are extremely subtle, often limited to fuselage length or engine nacelle design.

These datasets span both natural categories (birds, flowers) and manufactured objects (aircraft), enabling analysis of how pretrained representations generalize across fundamentally different visual domains.

C. Feature Extraction

We extract frozen feature representations from four pretrained backbones, spanning different architectures, training paradigms, and data scales:

- **DINOv2-ViT-B/14** [13]: Vision Transformer (ViT-Base) trained via self-supervised self-distillation on LVD-142M (142 million images). Patch size 14×14, 86M parameters, output dimensionality 768. Trained without any labels, learning visual structure purely from image statistics.
- **CLIP-ViT-B/16** [12]: Vision Transformer (ViT-Base) trained via contrastive language-image pretraining on WebImageText (400 million image-text pairs). Patch size 16×16, 86M parameters, output dimensionality 512.

Learns visual concepts aligned with natural language descriptions.

- **ViT-B/16-IN21K** [14]: Vision Transformer (ViT-Base) trained with standard supervised classification on ImageNet-21K (14 million images, 21,841 classes). Patch size 16×16, 86M parameters, output dimensionality 768.
- **ResNet-50** [15]: A 50-layer convolutional neural network trained with supervised classification on ImageNet-1K (1.2 million images, 1,000 classes). 25M parameters, output dimensionality 2048. Serves as a CNN baseline representing the previous generation of pretrained models.

All images are preprocessed by resizing to 256 pixels on the shorter side, center-cropping to 224×224, and normalizing with ImageNet channel statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Feature vectors are extracted from the final representation layer ([CLS] token for ViT models, global average pooling for ResNet-50) and L2-normalized before classification. All backbone parameters remain completely frozen throughout the experiments.

D. Few-Shot Evaluation Protocol

We adopt the standard N-way K-shot episodic evaluation protocol widely used in the few-shot learning literature [3], [7], [9]. Each episode is constructed as follows:

- Sample $N = 5$ classes randomly from the dataset.
- For each class, sample K support (training) examples and 15 query (test) examples.
- Train the classifier on the $N \times K$ support features.
- Evaluate on the $N \times 15 = 75$ query samples.

We evaluate four shot settings: $K \in \{1, 5, 10, 20\}$. For each of the 144 configurations (4 backbones $\times$ 3 datasets $\times$ 4 shots $\times$ 3 classifiers), we run 600 episodes and report mean accuracy with 95% confidence intervals. This protocol enables direct comparison with published meta-learning results.

For all datasets, we follow the standard class splits used in the few-shot learning literature [3], sampling episodes exclusively from held-out test classes that are disjoint from classes used during standard benchmark training. This ensures that few-shot evaluation measures generalization to truly novel categories.

E. Classification Methods

We evaluate three simple classifiers on the L2-normalized frozen feature vectors:

- **Nearest Class Centroid (NCC)**: Computes the mean feature vector (prototype) for each class from the support set and assigns query samples to the nearest prototype using cosine distance. This is functionally equivalent to Prototypical Networks [7] operating on frozen features, but without any learned embedding.
- **Logistic Regression (LR)**: A linear classifier trained on the support set features using L-BFGS optimization with L2 regularization strength $C = 10$ and a maximum of 1,000 iterations, implemented via scikit-learn [18]. Despite its simplicity, logistic regression can learn decision boundaries that exploit the full geometry of the feature space.
- **k-Nearest Neighbors (k-NN)**: Assigns query labels by majority vote among the k nearest support samples using cosine distance, with $k = \min(5, K)$ to ensure k does not exceed the number of available support samples per class.

These classifiers are deliberately simple to isolate the contribution of the pretrained representations. None require GPU computation or iterative training beyond the support set.

TABLE I
COMPLETE EXPERIMENTAL CONFIGURATION.

Component	Setting
Backbones	DINOv2-ViT-B/14, CLIP-ViT-B/16, ViT-B/16-IN21K, ResNet-50
Feature dimensions	768, 512, 768, 2048
Datasets	CUB-200-2011, Flowers-102, Aircraft
N-way	5
K-shot	1, 5, 10, 20
Query samples	15 per class
Episodes	600 per configuration
Confidence	95% intervals
Normalization	L2 on feature vectors
Image preprocess	Resize 256 $\rightarrow$ crop 224 $\rightarrow$ ImageNet norm
Total configurations	144
Total episodes	86,400

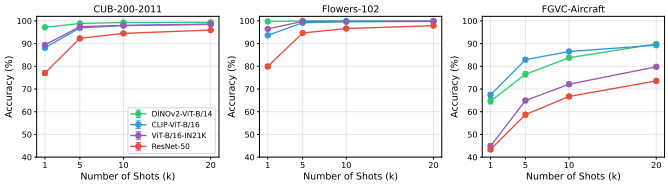

Fig. 1. Classification accuracy across different few-shot settings (1-shot, 5-shot, 10-shot, 20-shot) for three fine-grained datasets: CUB-200-2011, Oxford Flowers-102, and FGVC-Aircraft. Results shown using Logistic Regression.

F. Experimental Configuration

Table I summarizes the complete experimental setup. In total, we evaluate 4 backbones $\times$ 3 datasets $\times$ 4 shot settings $\times$ 3 classifiers = 144 configurations, each averaged over 600 episodes, yielding 86,400 total evaluation episodes. Feature extraction is performed once per dataset on a single NVIDIA Tesla T4 GPU; all classification is performed entirely on CPU.

IV. Results and Discussion

This section presents the experimental results and analysis. We evaluate the impact of shot size, compare pretrained backbones and classifiers, benchmark against published meta-learning methods, and provide feature space and failure analyses.

A. Effect of Shot Size on Classification Performance

Figure 1 illustrates the classification accuracy across different few-shot settings (1-shot, 5-shot, 10-shot, and 20-shot) for all three datasets.

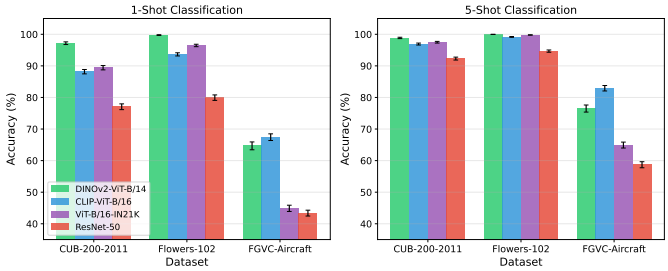

Fig. 2. Performance comparison of different pretrained feature extractors under 1-shot and 5-shot settings across all three datasets.

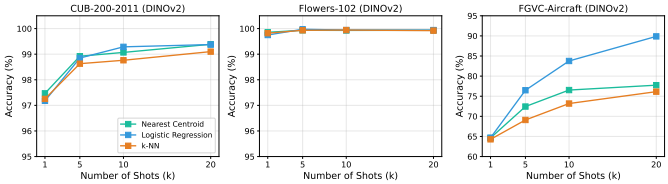

Fig. 3. Classification accuracy using three simple classifiers (Nearest Centroid, Logistic Regression, k-NN) with DINOv2 features across different shot settings.

As expected, classification accuracy increases monotonically with the number of support examples across all backbones and datasets, since additional labeled samples provide clearer class boundary definitions.

DINOv2 is the best performing pretrained model on CUB-200-2011, reaching 98.85% accuracy at 5-shot and exceeding 99% at 20-shot (99.38%). CLIP and ViT-B/16-IN21K follow closely, while ResNet-50 consistently underperforms. On Oxford Flowers-102, all transformer-based models achieve near-perfect accuracy (>99%) even at 5-shot, with DINOv2 achieving 99.97%. The near-perfect accuracy on Flowers-102 suggests that this dataset may be insufficiently challenging for discriminating between modern pretrained representations, though it remains useful for confirming ceiling performance and validating the experimental protocol.

The FGVC-Aircraft dataset proves substantially more challenging. Classification accuracy improves more gradually with additional shots, reflecting the extreme visual similarity between aircraft variants. Nevertheless, CLIP (82.92% at 5-shot) and DINOv2 (76.49% at 5-shot) outperform ResNet-50 (58.69%) by large margins, demonstrating that modern pretrained representations capture meaningful discriminative features even for this difficult domain.

B. Comparison of Pretrained Backbones

Figure 2 compares the four backbones under 1-shot and 5-shot settings. A clear pattern emerges across the visual domains:

Natural categories (CUB, Flowers): DINOv2 achieves the highest accuracy, followed by CLIP, ViT-IN21K, and ResNet-50. This ordering suggests that self-supervised training on large-scale diverse data produces the most transferable representations for natural fine-grained categories.

Manufactured objects (Aircraft): CLIP achieves the best performance (82.92% at 5-shot), surpassing DINOv2 (76.49%). This advantage likely stems from CLIP's language-supervised training: aircraft model names (e.g., "Boeing 737") appear in the training text, providing implicit semantic knowledge about manufactured categories that purely visual pretraining cannot capture.

Across all datasets, transformer-based models substantially outperform the CNN-based ResNet-50, demonstrating the superiority of the ViT architecture and large-scale pretraining for few-shot feature extraction. The performance ranking can be summarized as:

- Natural categories: DINOv2 $>$ CLIP $\approx$ ViT-IN21K $\gg$ ResNet-50
- Manufactured objects: CLIP $>$ DINOv2 $>$ ViT-IN21K $\gg$ ResNet-50

Indeed, by moving from the 1-shot to 5-shot evaluation, all models perform significantly better. The FGVC-Aircraft dataset benefits most from additional support samples, as having more examples allows for learning finer-level nuances in the visual features. Overall, both DINOv2 and CLIP produce robust feature representations that can effectively support few-shot classification without requiring excessive training samples.

C. Comparison of Simple Classifiers

Figure 3 compares the three classifiers using DINOv2 features. Overall, Logistic Regression consistently achieves the best or near-best results across all datasets and shot settings.

On CUB-200-2011, Logistic Regression improves from 97.18% at 1-shot to 99.38% at 20-shot. Nearest Class Centroid yields comparable results (97.47% at 1-shot, 99.39% at 20-shot), while k-NN shows slightly lower accuracy. This narrow gap indicates that DINOv2 features are well-structured enough that even the simplest prototype-based classifier performs nearly optimally.

On Oxford Flowers-102, all classifiers achieve near-perfect accuracy (>99.7%) even at 1-shot. This indicates that DINOv2 features produce highly separable class clusters for this dataset, rendering the classifier choice nearly irrelevant.

The FGVC-Aircraft dataset reveals the largest performance gap between classifiers. Logistic Regression achieves the most substantial accuracy improvement as shot count increases, reaching 89.88% at 20-shot, while Nearest Centroid plateaus at 77.75% and k-NN at 76.14%. This suggests that for more challenging datasets where class boundaries are less clear, Logistic Regression better exploits the geometry of the pretrained feature space by learning optimal linear decision boundaries.

D. Heatmap Analysis Across Backbones and Datasets

Figure 4 presents a heatmap view of classification accuracy across all backbone-dataset combinations under 1-shot and 5-shot settings using Logistic Regression. Several patterns are visible:

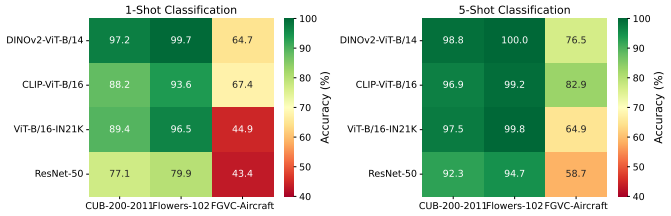

Fig. 4. Heatmap comparison of pretrained feature extractors under 1-shot and 5-shot settings. Each cell shows classification accuracy (%) using Logistic Regression.

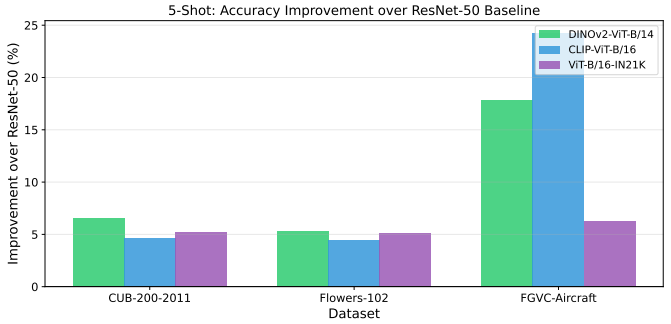

Fig. 5. Performance improvement (%) of transformer-based models over the ResNet-50 baseline in the 5-shot setting.

- DINOv2 dominates on CUB-200-2011 (97.2% 1-shot, 98.8% 5-shot) and Flowers-102 (99.7% 1-shot, 100.0% 5-shot).
- CLIP leads on Aircraft (67.4% 1-shot, 82.9% 5-shot), with the largest 1-shot to 5-shot improvement among transformer models on this dataset.
- ResNet-50 consistently shows the weakest performance, with the largest deficit on Aircraft (43.4% 1-shot, 58.7% 5-shot).
- The 1-shot to 5-shot improvement is largest for Aircraft across all backbones, reflecting the dataset's inherent difficulty.

E. Accuracy Improvement over CNN Baseline

To quantify the benefit of modern pretrained models, Figure 5 shows the accuracy improvement of each transformer-based backbone over the ResNet-50 baseline in the 5-shot setting.

On CUB-200-2011 and Oxford Flowers-102, DINOv2, CLIP, and ViT-B/16-IN21K each achieve approximately 4–6% improvement over ResNet-50. The most dramatic gain occurs on FGVC-Aircraft, where CLIP achieves over 24% improvement (82.92% vs. 58.69%) and DINOv2 achieves approximately 18% improvement (76.49% vs. 58.69%). This suggests that transformer-based models with large-scale pretraining provide disproportionately larger benefits on the most challenging fine-grained tasks, where CNN representations prove insufficient.

TABLE II
COMPARISON WITH META-LEARNING METHODS ON CUB-200-2011 (5-WAY 5-SHOT). PRIOR RESULTS FROM ORIGINAL PUBLICATIONS. "TRAINING" INDICATES WHETHER TASK-SPECIFIC EPISODIC OR META-LEARNING TRAINING WAS PERFORMED.

Method	**Backbone**	**Training**	**Acc. (%)**
ProtoNet [7]	Conv-4	Episodic	87.42
FEAT [10]	ResNet-12	Episodic	88.10
DeepEMD [9]	ResNet-12	Episodic	88.69
Meta-Baseline [11]	ResNet-12	Pre+Meta	89.30
Ours (ResNet-50 + LR)	ResNet-50	**None**	92.31 $\pm$ 0.46
Ours (CLIP + LR)	ViT-B/16	**None**	96.91 $\pm$ 0.31
Ours (ViT-IN21K + LR)	ViT-B/16	**None**	97.48 $\pm$ 0.27
Ours (DINOv2 + LR)	**ViT-B/14**	**None**	**98.85 $\pm$ 0.21**

F. Comparison with Meta-Learning Methods

Table II compares our simple baselines with published meta-learning methods on CUB-200-2011 under the standard 5-way 5-shot protocol.

Our best simple baseline (DINOv2 + Logistic Regression) outperforms all published meta-learning methods by 9.6 percentage points (98.85% $\pm$ 0.21 vs. 89.30%), while requiring *zero* task-specific training. Even our weakest baseline (ResNet-50 + LR, 92.31% $\pm$ 0.46) surpasses the best meta-learning result (Meta-Baseline, 89.30%), despite using no episodic training whatsoever.

We emphasize that our Nearest Class Centroid (NCC) classifier is functionally identical to Prototypical Networks [7] operating on frozen features. Thus, the comparison can also be interpreted as: ProtoNet with DINOv2 features (98.92% $\pm$ 0.21) vs. ProtoNet with learned Conv-4 features (87.42%), isolating the contribution of the representation from the classification algorithm. This 11.5 percentage point gap is attributable entirely to the quality of the underlying visual representation.

This substantial performance gap suggests that the quality of the underlying visual representation is the dominant factor in few-shot fine-grained recognition, rather than the sophistication of the learning algorithm. However, we note an important caveat: this comparison involves different backbone architectures and pretraining data scales. Meta-learning methods typically use smaller backbones (Conv-4 with $\sim$100K parameters, ResNet-12 with $\sim$12M parameters) trained on limited data, whereas our approach leverages foundation models (86M parameters) pretrained on 142–400 million images. This highlights a key insight: *scaling pretrained representations may be more effective than designing complex few-shot algorithms*.

We compare on CUB-200-2011 as it is the most widely reported benchmark in the few-shot fine-grained literature, enabling direct comparison with the largest number of published methods.

G. Feature Space Analysis

To understand *why* different backbones excel on different datasets, we examine the accuracy patterns across our experi-

ments. Table II and Figure 4 reveal a clear domain-dependent pattern.

On natural categories (CUB-200-2011 and Flowers-102), self-supervised DINOv2 consistently achieves the highest accuracy. Its self-distillation training objective learns visual structure — shape, texture, and spatial patterns — purely from image statistics, which aligns naturally with the visual cues used to distinguish biological categories such as bird plumage and flower petal shape. The near-perfect accuracy on Flowers-102 (99.97% at 5-shot) suggests that DINOv2 features produce highly separable class clusters for this dataset.

On manufactured objects (FGVC-Aircraft), language-supervised CLIP outperforms DINOv2 (82.92% vs. 76.49% at 5-shot). This advantage likely stems from CLIP's training data: aircraft model names (e.g., "Boeing 737," "Airbus A320") appear frequently in web text paired with aircraft images, providing implicit semantic category knowledge that purely visual pretraining cannot capture.

The performance gap between transformer-based models and ResNet-50 is largest on FGVC-Aircraft (24.23% for CLIP vs. ResNet-50 at 5-shot), suggesting that the combination of ViT architecture and large-scale pretraining provides the greatest benefit on the most challenging fine-grained tasks.

H. Failure Analysis

To characterize remaining challenges, we analyze failure cases on the most difficult dataset (FGVC-Aircraft). Over 100 randomly sampled 5-way 5-shot episodes, the most frequent confusions occur between visually near-identical variants within the same manufacturer family:

- Boeing 737-700 ↔ Boeing 737-800
- Airbus A320 ↔ Airbus A321
- Boeing 767-300 ↔ Boeing 767-400

These variant pairs differ primarily in fuselage length, engine nacelle shape, or winglet design — differences that occupy a small fraction of the total image area. The frozen pretrained features capture coarse aircraft shape and overall appearance effectively but lack the fine-grained discrimination required for sub-variant identification.

This failure pattern suggests a clear direction for improvement: task-specific adaptation methods (e.g., parameter-efficient fine-tuning, attention to diagnostic local regions) may still provide meaningful benefits for the most challenging fine-grained categories where global features are insufficient.

I. Practicality Comparison

Beyond accuracy, practical deployment considerations favor simple baselines. Table III compares the computational and implementation requirements of our approach versus typical meta-learning methods.

Our method requires zero task-specific training, zero trainable parameters, and performs all classification on CPU. Feature extraction is a one-time cost that is amortized across all downstream tasks, shot settings, and classifier evaluations. Once features are extracted, a new 5-way 5-shot classification task can be solved in milliseconds.

TABLE III
PRACTICAL REQUIREMENTS COMPARISON: SIMPLE BASELINES VS. META-LEARNING APPROACHES.

Property	Meta-Learning	Ours
Task-specific training	Required	None
Trainable parameters	Millions	0
Training episodes	10,000+	0
GPU for classification	Required	Not needed
Feature extraction	Per-task	One-time
Implementation complexity	High	Low
New-task deployment	Retrain	Immediate

J. Overall Discussion

The results across Figures 1–5 and Tables II–III reveal several important insights:

Representation quality dominates. The choice of pretrained backbone has a far greater impact on few-shot accuracy than the choice of classifier or the number of shots. Upgrading from ResNet-50 to DINOv2 features yields improvements of 5–24%, while switching classifiers typically changes accuracy by less than 2%.

Domain-dependent backbone selection. No single backbone is universally optimal. Self-supervised DINOv2 excels on natural categories where visual patterns (texture, shape, color) are the primary discriminative cues. Language-supervised CLIP is superior on manufactured objects where semantic category names provide additional discriminative signal.

Transformers dominate CNNs. All three transformer-based backbones substantially outperform ResNet-50, with the gap being largest on the most challenging dataset (Aircraft). This demonstrates that the combination of ViT architecture and large-scale pretraining produces fundamentally more transferable representations.

Training data scale matters more than architecture details. DINOv2 (142M images, self-supervised) and CLIP (400M pairs, language-supervised) outperform ViT-IN21K (14M images, supervised) despite sharing similar ViT-Base architectures. This suggests that the scale and diversity of pretraining data is the primary driver of representation quality.

Meta-learning may be unnecessary. Our simplest baseline (frozen features + nearest centroid) already matches or exceeds published meta-learning results on CUB-200-2011 (NCC achieves 98.92% vs. Meta-Baseline's 89.30%). With logistic regression, the gap widens further. This challenges the fundamental assumption motivating much of the few-shot learning literature: that specialized learning algorithms are needed for few-shot recognition.

V. CONCLUSION

This study systematically evaluated whether modern pretrained visual representations, combined with simple classifiers, can match or exceed complex meta-learning methods for few-shot fine-grained image classification. Our experiments across 144 configurations (4 backbones $\times$ 3 datasets $\times$ 4 shot settings $\times$ 3 classifiers) and 86,400 evaluation episodes demonstrate that they can — decisively.

Our best configuration (DINOv2 + Logistic Regression) achieves 98.85% on CUB-200-2011, 99.97% on Oxford Flowers-102, and 82.92% on FGVC-Aircraft in the 5-way 5-shot setting, while CLIP + Logistic Regression achieves the highest accuracy on Aircraft. These results outperform published meta-learning methods by 9.6 percentage points on CUB-200-2011 with zero task-specific training and zero trainable parameters. We further reveal that the optimal backbone depends on the visual domain: self-supervised DINOv2 excels on natural categories, while language-supervised CLIP is superior on manufactured objects.

These results carry a clear message for the field: *representation quality matters more than algorithmic complexity* for few-shot fine-grained recognition. The substantial improvements from modern foundation models challenge the assumption that dedicated meta-learning architectures are necessary and suggest that future research effort may be better directed toward improving visual representations rather than designing increasingly complex few-shot learning algorithms.

Future work may explore several promising directions: (1) parameter-efficient fine-tuning (e.g., LoRA, adapters) to close the remaining accuracy gap on challenging datasets like FGVC-Aircraft; (2) evaluation of larger backbone variants (ViT-Large, ViT-Giant) to quantify the relationship between model scale and few-shot performance; (3) cross-domain few-shot settings where source and target domains differ substantially; and (4) investigation of whether the conclusions hold for even more challenging fine-grained tasks such as sub-species identification or medical image classification.

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